

SSH Hardening & UFW Configuration

Objective

The objective of this task is to harden SSH access on the Ubuntu Server by replacing password-based authentication with SSH key based authentication. Disabling direct root login and configuring UFW to restrict SSH access to the trusted lab network. This configuration will only allow SSH access from the Kali Linux while preventing unauthorized SSH access from other networks.

Lab Environment

Server: Ubuntu Server

Client: Kali Linux

Server IP: 192.168.10.20

Kali Linux IP: 192.168.10.100

SSH Port: TCP/22

Firewall: UFW

SSH Authentication: Ed25519 key pair

SSH Key Pair Generation

An Ed25519 SSH key pair was generated on the kali Linux client.

The generated key pair consisted of:

Id_ed25519 ---- Private key Id_ed25519.pub --- Public key

The private key was retained on the kali system and the public key was deployed to the server.

```

(root@kali)-[/home/user]
# ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/root/.ssh/id_ed25519):
Enter passphrase for "/root/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_ed25519
Your public key has been saved in /root/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:wmpWN2jRKrSSsLllm0S+TmokSx5cJsV8yhV8G//wl34 root@kali
The key's randomart image is:
+--[ED25519 256]--+
|  o  ...          |
|    + + +         |
| . + = o =        |
| B B o = o        |
|+ @ o B S + .     |
|.X = = o . o o    |
|*. * +            |
|.* o              |
|o .               |
+-----[SHA256]-----+

(root@kali)-[/home/user]
# ls -la ~/.ssh
total 16
drwx----- 2 root root 4096 Aug 16 12:23 .
drwx----- 5 root root 4096 Aug 16 12:21 ..
-rw----- 1 root root 399 Aug 16 12:23 id_ed25519
-rw-r--r-- 1 root root 91 Aug 16 12:23 id_ed25519.pub

```

Public Key Deployment

The public key was copied from the Kali Linux client to the Ubuntu Server and added to the user's SSH authorized keys on the Ubuntu Server. This enables the server to authenticate the client using the corresponding private key instead of relying on password authentication.

```

(root@kali)-[/home/user]
# ssh-copy-id user@192.168.10.20
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are
already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to
install the new keys
user@192.168.10.20's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'user@192.168.10.20'"
and check to make sure that only the key(s) you wanted were added.

```

SSH Key-Based Authentication Testing

A new SSH connection was successfully established from Kali Linux to the Ubuntu Server after deploying the public key, which confirms the SSH key pair was functional before password authentication was disabled.

```
(root@kali)~[/home/user]
# ssh user@192.168.10.20
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-29-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Sun Aug 16 07:52:58 AM UTC 2026

System load:  0.0               Processes:    109
Usage of /:   14.0% of 24.44GB   Users logged in: 0
Memory usage: 14%              IPv4 address for enp0s3: 192.168.10.20
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

51 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Aug 14 07:41:11 2026 from 192.168.10.100
user@Ubuntu-Server:~$
```

Disabling Root login & Password Authentication

The Following settings were configured:

PermitRootLogin: no (prevents Direct ssh login using root account)

PasswordAuthentication: no (disables password based SSH authentication)

PermitEmptyPasswords: no

The SSH configuration was validated before applying the changes:

sudo sshd -t

The SSH service was then restarted to apply the configuration.

sudo systemctl restart ssh

```
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to "no" here!
PasswordAuthentication no
PermitEmptyPasswords no

# Change to "yes" to enable keyboard-interactive authentication. Depending on
# the system's configuration, this may involve passwords, challenge-response,
# one-time passwords or some combination of these and other methods.
# Beware issues with some PAM modules and threads.
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no

# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing. If this is enabled, PAM authentication will

user@Ubuntu-Server:~$ sudo sshd -t
user@Ubuntu-Server:~$ sudo systemctl restart ssh
user@Ubuntu-Server:~$
```

UFW Configuration

UFW was configured using a default-deny policy for incoming connections and a default-allow policy for outgoing connections:

sudo ufw default deny incoming

sudo ufw default allow outgoing

This prevents unsolicited incoming connections unless an explicit firewall rule allows them.

SSH Access Restriction to Lab Network

SSH access was restricted to the lab network 192.168.10.0/24.

The following rule was configured:

`sudo ufw allow from 192.168.10.0/24 to any port 22 proto tcp`

UFW was then enabled: `sudo ufw enable`

The resulting firewall configuration was verified using:

`sudo ufw status verbose`

The firewall showed TCP port 22 allowed only from 192.168.10.0/24.

```
user@Ubuntu-Server:~$ sudo ufw status verbose
[sudo: authenticate] Password:
Status: inactive
user@Ubuntu-Server:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
user@Ubuntu-Server:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
user@Ubuntu-Server:~$ sudo ufw allow from 192.168.10.0/24 to any port 22 proto tcp
WARN: Rule changed after normalization
Rules updated
user@Ubuntu-Server:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y/n)? y
Firewall is active and enabled on system startup
user@Ubuntu-Server:~$ sudo ufw status numbered
Status: active

      To      Action      From
      --      -
[ 1] 22/tcp    ALLOW IN    192.168.10.0/24

user@Ubuntu-Server:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

      To      Action      From
      --      -
22/tcp    ALLOW IN    192.168.10.0/24
```

Security Validation / Testing

The configuration was tested after applying the SSH and firewall hardening measures.

The SSH connection from the authorized lab network remained functional using the SSH key pair.

The UFW status confirmed that:

- Incoming traffic is denied by default.
- Outgoing traffic is allowed by default.
- TCP port 22 is allowed only from 192.168.10.0/24.
- No unrestricted SSH rule was present.

This provides two layers of protection for SSH:

1. **SSH authentication hardening** prevents password-based and direct root authentication.
2. **Host-based firewall filtering** restricts which network sources can reach the SSH service.

Conclusion

SSH on the Ubuntu Server was hardened by implementing key-based authentication, disabling root and password-based SSH access, and restricting SSH network access through UFW.

The resulting configuration follows the principle of least privilege by allowing only the required SSH service and limiting access to the trusted laboratory network.